AI Forward Deployed Engineer

The AI Architect & FDE Course



Week 3 Day 1

3 weeks to Forward Deployed Engineer

Theory Practical Projects

Week 3: Become an AI FDE

- AI Solutions Architect
- Security, Guardrails, API Gateways
- Enterprise Context Layer & Integrations
- Production Deployment
- Capstone

Week 2: Become an AI Architect

AI ARCHITECT

- Architecture, Evals, Observability
- Agentic RAG, MCP, Memory
- Durable State & Human in the Loop
- Agent Teams, Swarms, Fleet
- Self-Improvement, RSI

Week 1: Become an AI Engineer

AI ENGINEER

- Path to FDE
- Frontier LLMs & Agents
- Coding Agents & ADLC
- Open-Source, Local, Token Cost
- Harness Engineering

Taking stock

You can already:

- ✓ Deliver solutions that measurably solve business problems with evals and observability
- ✓ Build agents with memory using Agentic RAG and MCP
- ✓ Create durable and scalable agents in LangGraph with a human in the loop
- ✓ Engineer an agent harness that can improve itself at a business task

By the end of today, you'll be able to:

- ○ Detail the lifecycle for developing and deploying agents as an FDE
- ○ Outline the major systems architecture decisions for an FDE
- ○ Understand governance models for AI projects and responsible AI concerns
- ○ Explain the importance of handover, adoption and feedback for an FDE

The 2 meanings of ADLC

Quite confusing

TODAY



1. Building agents with code

The emerging process for building AI agents including risks & controls, evaluations, adoption, observability and maintenance.

LangChain: Agent Development Lifecycle · Microsoft Foundry: agent lifecycle

WEEK 1 DAY 3



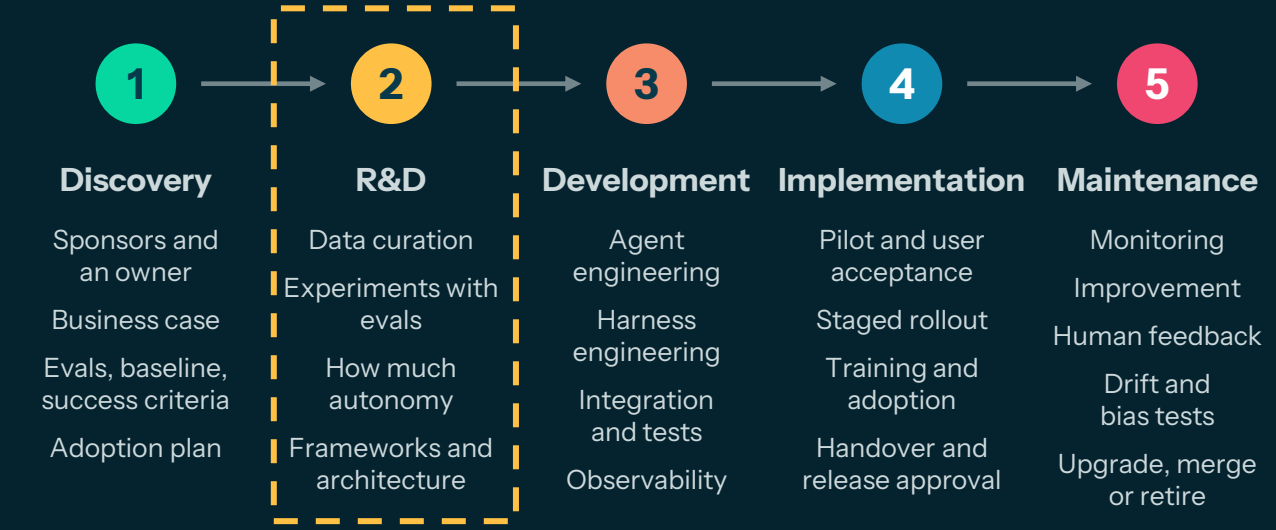
2. Building code with agents

The emerging process for writing software with agents like Claude Code while staying accountable for the results.

AWS: AI-Driven Development Life Cycle · GitHub Spec Kit SDD

The agent development lifecycle

This is an evolving field but these are emerging common phases



Ownership, governance, evals and change management throughout

The FDE through the lifecycle

From discovery at the client to feedback for your employer

Discovery → **Discovery and ideation**
Use cases, a baseline and success criteria

R&D → **Data curation and prototyping**
Data quality, an evaluation set and a prototype

Development → **Building with the client's engineers**
Integrations, guardrails, evals and observability

Implementation → **Deployment, handover and adoption**
A pilot, training, a runbook and a service owner

Maintenance → **Feedback to your employer**
What the product lacked and what to productize

More ideation

Discovery and ideation

Prioritize use cases that solve a measurable business problem

The bar has risen



Automation
Do things cheaper
"File my expenses"



Acceleration
Do things faster
"Get new clients"



Differentiation
Do different things
"Find new product strategy"

Three questions for each use case

1. Does it avoid the traps from week 2
2. Does it use your own data, connections and evals?
3. Is it automation, acceleration or differentiation?

Systems architecture

The systems around the agent



Week 3 Day 2

LLM and MCP gateways

One place to route model calls and control each agent's tools



Week 3 Day 3

Common integrations

Systems of record like CRM, claims and ERP, plus documents, email and tickets



Week 3 Day 3

The context and data layer

What the data means, which sources to trust, what the agent may see and its memory

Start as simple as possible



No major internal build



Keep prompts with the applications



No heavyweight, opinionated product



Expect a redesign within a year

Where the agent runs

3 ways to host an agent in production

	Examples	What you get
Your own infrastructure	**AWS Lambda** ECS, Cloud Run or Kubernetes Render	Any framework and model State and scaling are yours
A managed runtime	**AWS AgentCore Runtime** Google Agent Runtime Microsoft Foundry LangSmith Deployment	Any framework and model Memory and identity built in
The model lab runs the agent	Claude Managed Agents OpenAI Agents API	The lab's own harness Only the lab's models

Who decides

Three levels of governance and a tollgate at the end of each phase

Firmwide Leadership Monthly / Quarterly	Portfolio, policy and risk tolerance Executives across the business
Project committee Monthly	Scope, risk and releases for one project Sponsor, owner, security, audit and operations
Working group Weekly	Build, evaluate and run the agent Product, engineering, the business and the FDE

A sponsor signs off at the end of each phase



How much autonomy

Setting the right level of autonomy is a risk-based decision

Assess each action for **risk appetite**, **consequence**, **verifiability**, **reversibility** and **security, privacy and IP**.



Automatic

Read the claim
Search the wording
Calculate the payout
Approve, decline or refer
Pay up to $1,000

Approval first

Pay over $1,000
Edit the prompt or rules
Change the model
Edit the recipe

Prohibited

Sign off its own payout
Change the $1,000 limit
See held-out answers
Edit the evaluation sets

For the human in the loop

A review screen
The evidence with buttons to approve, edit or reject

An audit trail
Every step and who approved each payout

A plan for delays
Timeouts, escalation and no double payments

Managing many agents

Some organizations manage a roster of agents like staff

Agents run like staff

Microsoft Agent 365 and Entra Agent ID
A registry of agents and their identities

Workday Agent System of Record
Agents managed beside employees

AWS Agent Registry
A governed catalog of agents

OpenAI Frontier
“AI coworkers” with identities

UnderCover’s agents

Features of one product
Dispatcher, claim agents and improver

Accountable owner
The owner of the claims product

Versions and permissions
A handler approves every change

Traces and spend
Recorded for every claim and run

My recommendation: treat agents like any other software asset that solves a commercial problem.

Responsible AI

6 pillars of responsible AI to be incorporated in the plan


Safety


Fairness / Bias


Security


Privacy


Transparency


Accountability

Most important is to budget and plan for these activities during the project and after implementation in prod

Privacy

When using LLM APIs or Products:



Prompts are sent to the external company

Everything in the context is sent to the model provider



Know the training policy

Consumer chat products may train on conversations;
business APIs typically do not



Minimize personal data

Send only what the task needs; redact or anonymize the rest as appropriate

Treat the context window like data sent to any third-party;
but no worse than data sent to a third party…

If you'd normally trust a SOC2 certified vendor with data,
then a SOC2 certified LLM provider is the same!

Accountability



A human is always accountable for the quality and the ultimate outcomes

What this means in practice



A human in the loop wherever appropriate

Approval steps for consequential actions



An audit trail, alerting, flagging

Monitor and quickly correct issues



Clear ownership

The LLM is the tool, not the owner

Regulation is arriving like the EU AI Act

Handover and adoption

The client's team can run it without the FDE

Handover



Ownership and escalation



Knowledge transfer



Ongoing measurement

Adoption



Training and support



Sponsors and top-down change



Demonstrate output accuracy



Transparency on usage

Today's lab

Setting up AWS for Week 3

1. **Open an AWS account**
   The Paid plan with a card. New customers get up to $200 in credits.

2. **Turn on MFA for root**
   A passkey or an authenticator app on the root user

3. **Choose `us-east-1`**
   Set the console's region selector and leave it there

4. **Billing access and a budget**
   $10 a month with alerts at 50, 80 and 100%, credits excluded

5. **Create the IAM user**
   The `undercover-course` policy and group, then an access key

6. **Install the AWS CLI v2**

```
aws configure
aws sts get-caller-identity
```

Nothing on AgentCore today
The Gateway and the Policy come on Day 2

The free route
A Render account with written instructions

One region for everyone

We build everything for Week 3 in one AWS region

`us-east-1`

US East (N. Virginia)
For students in India and Europe too



Every feature we use
Runtime, Gateway, Identity, Policy, Memory and Observability



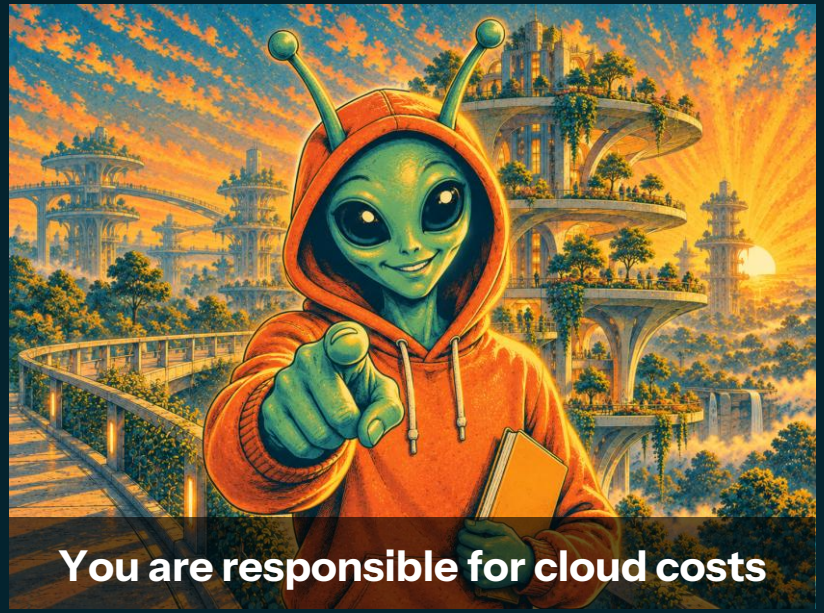
Gaps in Memory
Missing from six regions and used on Week 3 Day 3



The same as mine
Console paths, screenshots and the CDK bootstrap

Missing resources? Check the console's region selector

AWS and API costs



AWS is a commercial platform

Usage beyond your credits goes on your card

Set up a budget and monitor it

Turn on email alerts and check your costs often

Not comfortable with that?

Take the Render route, or just watch me on AWS

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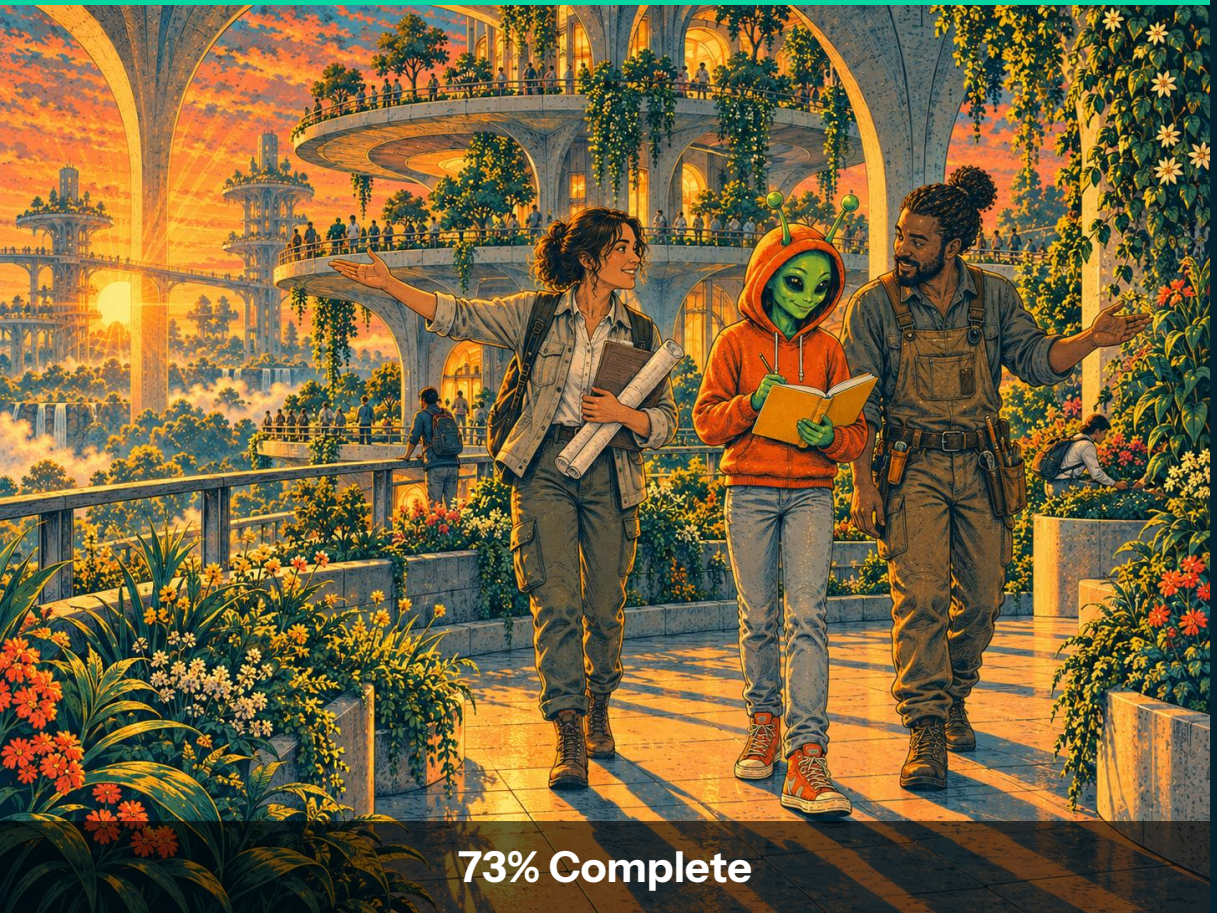
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73% Complete